

Oecologia Decision for OECO-D-26-00039 - [EMID:b03f628d3497e061]

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Carlos L. Ballare <em@editorialmanager.com>

Di 14.04.2026 16:40

An: Thomas, Daniel Churchill <Daniel.Thomas@uni-bayreuth.de>;

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"Frederic J. Pascual Dose" f.pascualdose@stud.uni-goettingen.de, "Jona Motzkus" jona.motzkus@uni-bayreuth.de

Re: "Scaling up by looking closer: extending the range of moss-Nostoc symbioses deeper into central Europe and to new moss hosts in the forest canopy" (Ms. No. %MS_NUMBER %)

Decision: Major Revision

Dear Dr. Thomas,

Your manuscript has been reviewed for Oecologia and one of our editors, Dr Evan DeLucia, has made a recommendation of Major Revision. I concur with this recommendation and believe that a major revision of this manuscript could produce an acceptable paper. As you will see below, the expert reviewers provided detailed and constructive comments, which are well summarized in the handling editor's report.

If you wish to revise your manuscript for further consideration, please address the points raised in the reviews and submit your revision by 13 Jun 2026

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at <https://www.editorialmanager.com/oeco/>. In your response letter, please indicate how you have addressed each individual comment. Of course, if you disagree with specific comments raised in the reviews, feel free to respond accordingly. The original handling editor will be notified of your revision and will continue to handle your manuscript. A final decision on the suitability of this submission for publication in Oecologia will be made on the basis of the revised manuscript and your response to the referees.

Please review our detailed author instructions for proper manuscript formatting: "Oecologia Author Instructions General" (Manuscript guidelines) and "Oecologia Author Instructions Figures" (Artwork instructions). These files are located at:

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and

http://www.springer.com/cda/content/document/cda_downloadaddocument/Oecologia_Author_Instructions_Figures.pdf?SGWID=0-0-45-1135642-p1081245

My assistant has reviewed your paper for formatting issues, particularly regarding the figures, and provided the following comments:

"Please, consider increasing the font size for the lettering of figures 9 and 10 (all letters and numbers in the figures should be at least 2 mm high at the intended final size of the figure in the printed article)."

Please make sure to submit editable source files (e.g. Word). We will return manuscripts which do not follow these guidelines.

Thank you for considering *Oecologia* as an outlet for your research results. We look forward to receiving your revised manuscript. If you choose not to undertake these revisions, then I wish you the best of luck in finding a more suitable outlet for your work.

Sincerely,

Carlos L. Ballaré
Editor in Chief, Plant-Animal, Plant-Microbe Interactions
Oecologia

Comments to Authors from Handling Editor and Reviewers

I agree with the external reviewers that this manuscript presents novel and interesting data and can potentially increase our understanding of the role of nitrogen fixation in forests. I encourage the authors to carefully revise their manuscript paying close attention to the suggestions provided by the external reviewers. Please pay careful attention to the comments provided by Reviewer 1, particularly the request to tighten up the language and more clearly justify the study.

Reviewer #1: In this manuscript, the authors describe a study in which they looked for colonies from nostocacean, nitrogen-fixing cyanobacteria in five moss species sampled in the Franconia region of central Europe, using microscopy, culturing and DNA sequencing. They have concluded that moss-cyanobacteria symbioses, which are known to be important in boreal forest and arctic tundra ecosystems, have a larger range of hosts and locations than previously thought. This makes the study quite interesting and timely, since it expands the potential target areas for studies in this topic.

However, the paper has lots of formatting, typing and informal language issues that should be revised. Those should be quite easy to fix. However, there are several other problems with the current version of the manuscript that must be fixed before publication. The objectives and hypotheses of the study are not clear in the Introduction or elsewhere, which makes it hard to understand the reasons behind the experimental design. For example, although they have collected five different moss species from different locations, they've only sequenced *nifH* genes associated with one moss species, and apparently without doing any replication. The Methods section is confusing, since it follows an unusual structure and goes back and forth on several procedures, making it unclear what was done and why. The parts describing the DNA sequencing are especially hard to follow. Furthermore, although it's a bit repetitive, the Methods section still lacks fundamental information, which harms the reproducibility of the study. The Discussion section could use many more references, and some of the conclusions should be reconsidered due to lack of evidence.

Despite these flaws, I think the manuscript could still provide an interesting contribution to the field after a major revision. So here follow some more specific comments on parts of the manuscript that I believe are unclear and should be improved.

*Abstract

Line 7, "Nostoc-feather-moss": The second hyphen here ("feather-moss") is unnecessary and makes this a bit confusing for readers who are not familiar with mosses. I advise you to remove the hyphen to improve the readability of the paper.

*Introduction

Line 17 and elsewhere, "*nifH*": Don't forget to italicize the gene names.

Line 46, "*Sphagnum*"; Lines 54, 56, 58 and others throughout the manuscript — "*Nostoc*": Please don't forget to italicize

genus and species names either.

Lines 66-67, "fixed organic nitrogen in the form of fixed, organic nitrogen": This sentence is redundant, so could you please rephrase it?

Line 74: This figure legend is super short and not very informative, so there's room to improve it. What do "o", "m" and "u" mean? What are the circles representing? Where exactly are the Nostoc colonies? Does (c) mean ©?

*Methods

Line 108, "They were determined using standard literature": It's a bit unclear to me what you mean by determined and standard literature. Did you mean that the mosses were taxonomically identified using this reference work?

Line 117: "Bei" is not part of the location name, right? Can it be translated to "near"? Also, how many samples of each species were taken? Were there any replicates?

Lines 118, 119, 128: The formatting of the titles and subtitles of these subsections made the structure of the methods quite confusing. It looks like Microbiology is another section at the same level as Methods instead of a subsection. Please review this to make sure it properly follows Oecologia's article format.

Lines 120-122: Which equipment did you use for this?

Line 123, "amplification of NifH and 16S rRNA marker genes": A bit of a small detail, but NifH refers to the protein, not to the marker gene.

Lines 122-124: How was PCR performed?

Lines 124-125: How was DNA extraction performed?

Lines 125-127: How was surface sterilization performed?

Lines 130-131: Please add manufacturer and model information for the microscopes used.

Lines 131-133: How was DNA purification and PCR amplification performed?

Lines 133-135, "Sectioning was done using sections of one centimeter from the top (o), middle (m) and bottom (u) of caulids": How did you do the sections? Did you just use e.g. a scalpel or was another type of instrument necessary?

Lines 140-141, "eukaryotic microalgae": There are no prokaryotic microalgae, so this term is redundant. Microalgae is sufficient here.

Lines 139-152-142: This long paragraph seems more appropriate for the Introduction or Discussion sections.

Line 144-145, "To test the hypothesis that cyanobacteria inhabit endophytic compartments of the host mosses, some Hylocomium moss samples were surface sterilized" — It would be easier to follow the reasoning behind your methods if you discussed these and other hypotheses that guided the design of your study in the introduction more explicitly. But, as far as I know, feather mosses like Hylocomium splendens do not have proper endophytic compartments due to their simple anatomy. As non-vascular plants, they don't have xylems or phloems, and their organs are quite basic, with tissues composed of a single cell layer. This is different than the case of Sphagnum, for example, which has empty cells filled with water (hyalocytes) that can be colonized by microbes and could indeed be considered endophytic compartments. So would you expect to find bacteria inside stems or leaves, for example? It doesn't seem like there'd be room for them.

Lines 153-154, "Surface sterilized samples were then again inspected for visible cyanobacterial colonization": I understand that several papers claim to have sterilized the surface of mosses and other plants, but seldom do they provide evidence for that. You mentioned in the introduction that cyanobacteria "appear to prefer to epiphytically inhabit the spaces in protected leaflet-axils along moss caulids ('stems'), or in incurved regions of leaflet-tissue and other protected areas"

(Lines 56-58), so it should be quite hard to sterilize the surface of the moss. Plus, you've only checked for the presence of cyanobacteria, not other microbes. I understand what you were trying to achieve here, but I think you should rephrase this section to more accurately represent what you intended to do.

Line 155, "Nostoc-like organisms": Throughout the manuscript, some times you refer to these bacteria as "Nostoc" and other times as "Nostoc-like". Considering that Nostoc is indeed a polyphyletic genus and that you show another proof of that in your own Results section, I think it's more appropriate to use "Nostoc-like" along your manuscript and to highlight the interesting underestimated diversity of the cyanobacteria in association with mosses. You could also use nostocacean instead if you prefer to refer to their family instead of the genus.

Line 160, "filamentous bacteria that were observed in sheathed syncoenobia": Sorry, but I've never heard the term syncoenobia used when referring to cyanobacteria. Are you sure that terminology is used in this context? Could you clarify what it means here?

Line 161, "hetercytes": There's a typo here. It should read heterocytes.

Line 161 and elsewhere, "BG-110": Another small detail here, but could you please make the '0' here a subscript character? Or else it reads like 110, which is inaccurate.

Lines 167 and 170, "eukaryotic algae": Here too "eukaryotic" is redundant. Please just use "algae".

Lines 176-177, "As of time of writing": I don't think you need to say this here.

Lines 177-178, "Nostoc-like organisms were successfully isolated and cultivated from 4 separate moss host species": But in Table 1 you show information for 5 moss species. What happened to the fifth one?

Line 185, "Anabaena variabilis": This species has been reclassified in the genus Trichormus by Komarek and Anagnostidis in 1989. Please see <http://www.cyanodb.cz/#/Trichormus>.

Lines 185 and 188: How were these plants surface-sterilized?

Line 196, NifH genes: Here too, NifH refers to the protein, while nifH refers to the gene.

Lines 196-198, "NifH genes from DNA of whole-community moss+microbiome were amplified from samples of H. splendens collected from the University of Bayreuth Ecological Botanical Garden": How many samples? And why was this only done for these samples? What was the objective here?

Lines 207-210: This is a bit confusing because later you don't mention which primers you used in this reaction, but in the next paragraph you mention a PCR of the 16S rRNA gene. So do these lines refer to nifH amplification in the Nostoc-like colonies? Please make this clear here.

Lines 215-216: The primer sequences are conventionally written as upper case characters.

Lines 223-225: Only the 16S rRNA gene amplicons were purified? What about the nifH amplicons?

Lines 231-233, "SQK-LSK114 kit was used in the case of 16S rRNA gene of Hylocomium-associated Nostoc, and the remaining 6 colony-PCR products were individually barcoded, multiplexed and sequenced jointly using a SQK-RBK114.24 kit": Sorry, but this is unclear to me because you're using different terms from the previous paragraphs. So Nanopore sequencing was not only done for colony PCR products like you stated in Line 227, but also for what you called "whole community moss+microbiome"? And you sequenced one Nostoc-like colony from one moss collected in each of the six locations in Table 1?

Lines 233-234, "In the first sequencing run, with only Hylocomium-associated Nostoc": Here, do you mean the two Hylocomium samples you had or just one of them? And if it's the latter, was it for the colony PCR or the moss+microbiome amplicons?

Line 254, "we used the BLAST-suite tools, specifically blastn and blastx programs": What version did you use?

Line 256, "nifH amino acid sequences"; Line 257, "our custom database of nifH sequences was a concatenation of NifH sequences": Please make it clearer when are you referring to DNA sequences (i.e., nifH) and protein sequences (i.e., NifH) by using their correct names. It seems like you're using nifH and NifH interchangeably and this is very confusing.

Line 261, "CIAlign": What version did you use?

Line 298, "Nostoc azollae (NC_014248.1)": This species was also reclassified in the genus Trichormus by Komarek and Anagnostidis in 1989.

*Results

Line 324, "Out of 5 observed Nostoc-like organisms": But you did observe only five. In Figures 2-7, you show much more than five colonies.

Lines 328-331, "The Pseudoamblystegium-associated Nostoc-like organism (...) was overtaken by an unidentified, co-cultured non-filamentous cyanobacterium (...), though some 16s data from the Nostoc-like organism were recovered": In your methods, you did not list chimera checks. So how can you ensure this sequence was legitimate?

Line 331, "some 16s data": Please avoid using informal language in a scientific paper such as this one. This should be corrected to "some 16S rRNA gene sequence data".

Lines 334-335, "Culture-free amplification of cyanobacterial nifH from Hylocomium moss tissue yielded two size classes of non-chimeric amplicons": Did you do any chimera check for these sequences? It's not listed in the Methods section. Furthermore, as far as I know, there's no way to tell if gel bands belong to chimeric sequences or not.

Lines 333-341: So the samples that you called moss+microbiomes were not sequenced, but merely PCR amplified? I'm sorry, but I really don't believe this experiment is useful for your study. Firstly, PCR of the nifH gene is notoriously difficult due to the high probability of non-specific amplification of homologous genes involved in photosynthetic processes like bchL/bchX/chlL, especially in plant-associated communities such as the ones you studied. Therefore, such work requires heavy filtering of sequencing data to select the target sequences. Secondly, determining amplicon size based on comparisons of gel bands to molecular ladders is highly inaccurate and should never be used to identify specific target products with so many bands. Thirdly, you chose the most studied (and therefore, less interesting) moss species among the ones you sampled for this experiment. It shouldn't have been difficult to do a similar procedure for the other, more uncommon mosses, and this would've been much more exciting. Lastly, there's no replication, so you can't even say much about how representative this result is. It's not clear to me what you intended with this, but regardless of that I strongly advise you to remove this part from the article.

Lines 350-352, "In Zygodon- and Homalia-associated cultures each appeared to host 3 alternative nitrogenases, each falling into a separate clade on the Kubota-Ininbergs nifH tree": You assume that you were able to get single strains via pipetting, but it's still possible that you had actually collected a few different ones belonging to different nostocacean cyanobacteria colonizing the same moss, isn't it? Unless you do genome sequencing, it's not yet possible to determine this. Correct if I'm wrong, but I don't remember seeing many reports of cyanobacteria with three alternative nitrogenases in literature.

Figure 9: You placed references instead of species names in several branches of the tree. That's not so informative. Could you please provide the species instead? Also, the writing in the figure is very small and hard to see. Is it possible to increase it?

Figure 9, "nifH gene phylogeny of select members of Nostocaceae": In the Methods, you mentioned you used amino acid sequences, not gene sequences for this tree. So which information is the correct one?

Table 2: Some extra information about how similar these sequences are would be welcome here. You could add sequence

identity and coverage based on BLAST alignments, for example.

Figure 10: This figure is even harder to read than the previous one. Can you try improving it? Also, you forgot to mention what the numbers in the nodes represent (bootstrap, perhaps?).

*Discussion

Lines 389-391, "Given the documented importance of moss-hosted cyanobacterial nitrogen in other systems (discussed above), the contribution of these symbioses to the primary productivity of montane forest ecosystems of our region of Central Europe is also likely to be significant": This would depend on basically three factors: 1) How strongly colonized by nitrogen-fixing cyanobacteria these mosses are; 2) How much nitrogen fixation these cyanobacteria could potentially carry out; and 3) How widespread are these mosses. I don't think you can say anything about the first two since you didn't quantify cyanobacterial colonization or measure nitrogen fixation, but I assume there are surveys in the region that point out how common these mosses are or maybe even estimate their biomass is. Could you add some information about this?

Line 396, "mono-algal": You meant mono-cyanobacterial, right?

Lines 403-404, "As such, the organisms cultured here likely greatly under-represents the diversity of Nostoc-like organisms in mosses in our region": That's possible, but it's also possible that nitrogen-fixing cyanobacteria from other families are present. You should not restrict it to nostocacean cyanobacteria.

Lines 429-431: Do you have a reference for that information?

Lines 470-471, "the Nostoc strain cultured from Homalia trichomanoides, closely related to Nostoc pruniforme SAG 62.79 (Fig. 10), has a version of nifH that falls within this "Stigonema" cluster": Like I said above, it's possible that Stigonema cells were present in the sequenced sample?

Line 474, "We also observed the possible endophytic colonization of moss stems by cyanobacteria": That's quite unlikely, considering how anatomically simple moss stems are. An insufficient surface sterilization procedure is a more likely explanation for what's seen in the microscopy pictures. I think you need to discuss this from a moss structural perspective, as well as providing additional evidence for it.

Line 480, "Stems ('caulids') and rhizoids of mosses are multicellular": What do you mean? All moss organs are multicellular.

Reviewer #2: This manuscript reports on the results of a survey on symbiotic relationships between Nostoc and epiphytic mosses in south-east Germany. One of the main findings is that symbiotic relationship with Nostoc was identified in two non-feather moss species, Orthotrichum and Zygodon. Surprisingly, the 16S sequence alignments did not yield a clear assignment to individual symbiotic clades within the Nostocaceae phylogeny. It is therefore unclear how much Nostoc contributes to nitrogen fixation in these mosses. The combination of Nostoc cultivation and microscopy with sequencing of nifH and 16S rRNA genes is an interesting approach for detecting Nostoc symbiosis in moss tissues. Results are presented clearly; microscopic images and figures are of high quality. The study is robust and provides novel insights into potential nitrogen fixation in epiphytic mosses in the canopy of temperate forests. Overall, the manuscript is written in a clear and convincing manner. After minor revisions, I can therefore recommend this manuscript for publication in Oecologia.

Specific comments

Title: The title is very long and not particularly catchy.

Abstract

L 6-8: Can you provide a rationale for this study - was it conducted solely out of interest in the symbiosis between Nostoc and mosses?

L 11-12: It should be noted that the mosses were sampled from the forest canopy (see title). What trees species?

L 17: What does 'single Nostoc-like organisms' mean? Does this refer to non-Nostoc diazotrophs? This should be clarified.

Introduction

L 66-67: the sentence should be rephrased: '... involving a transfer of fixed organic nitrogen in the form of fixed, organic nitrogen from heterocystous Nostoc colonies to the host plant...'

L93-95: This sentence could be deleted. Specific sampling regions and locations could be given in Material and Methods.

L100-101: Because of atmospheric deposition, nitrogen availability has been relatively high in German forests for many decades, which has often led to eutrophication and other negative effects. The question is rather whether nitrogen fixation occurs despite the high N deposition rate.

Discussion

L385-387: Reference is needed.

L389-393: For such statements, direct measurements of nitrogen fixation and moss biomass in the forests would be necessary. It can be assumed that mosses play only a minor role in supplying nitrogen to forests in Central Europe. However, symbiosis with diazotrophs is likely an important factor in the mosses' own nitrogen supply.

L434-435: Reference is needed.

L453-460: These statements are somewhat speculative, as there is no evidence of an Mo limitation in the canopy. A deficiency in other nutrients or micronutrients might play a greater role than a deficiency in Mo.

Conclusions

The conclusion is a bit too long-winded (in particular, L. 509-520) and could be summarized more concisely.

L508-509: N limitation, s. comment above

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